



## PROBLEM STATEMENT

### Advanced RAG Knowledge Base – AI Chatbot for Complex Banking Document Retrieval

| Topic                                                                           | Brief Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Suggested Technologies                                                                                                                                                                                                                                                                                                                                                                                      | Key Deliverables                                                                                                                                                                                                                                                                                    | Benefits to the Bank                                                                                                                                                                                                                                                          | Why This Problem Matters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| Advanced RAG Knowledge Base – AI Chatbot for Complex Banking Document Retrieval | <p>An advanced Retrieval-Augmented Generation (RAG) system that enables employees and customers to search across the bank's entire document repository (internal policies, regulations, circulars, operating procedures, contract templates, etc.) using natural language.</p> <p>Key differentiator:</p> <ul style="list-style-type: none"> <li>• Cross-references: Automatically follows and synthesizes referenced documents.</li> <li>• Amendments: Always applies the latest effective version of amended regulations.</li> <li>• Partial supersession: Excludes superseded clauses from responses.</li> <li>• Conflicting regulations: Detects inconsistencies and warns users.</li> </ul> | <ul style="list-style-type: none"> <li>• LLM (GPT-4, Claude, or equivalent)</li> <li>• Advanced RAG Pipeline</li> <li>• Knowledge Graph (Neo4j / NetworkX)</li> <li>• Versioning engine</li> <li>• Hybrid Search (Vector + BM25 + Graph Traversal)</li> <li>• Chunk-level supersession tracking</li> <li>• Vietnamese embeddings (PhoBERT, multilingual-e5)</li> <li>• FastAPI + Streamlit/React</li> </ul> | <ul style="list-style-type: none"> <li>• AI chatbot with source-cited answers</li> <li>• Knowledge graph visualization</li> <li>• Clause version timeline</li> <li>• Conflict detector</li> <li>• Admin dashboard for document updates</li> <li>• Benchmark comparison with standard RAG</li> </ul> | <ul style="list-style-type: none"> <li>• Ensures employees use current regulations</li> <li>• Reduces legal and compliance risks</li> <li>• Saves 2–3 hours/day for compliance teams</li> <li>• Faster onboarding</li> <li>• Standardized organizational knowledge</li> </ul> | <p>SHB manages thousands of internal and external regulatory documents (SBV, Government, Basel, etc.). A single regulation may be amended multiple times, while individual clauses can be partially superseded. Conventional RAG systems cannot accurately model these relationships, leading to answers based on outdated regulations and creating significant compliance risks. This is a challenging real-world AI problem that remains difficult even for many leading global financial institutions.</p> |

